

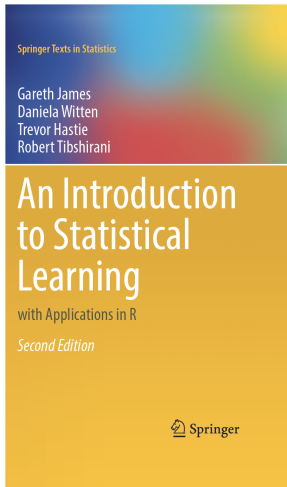
ML101
Lecture 03: Linear Regression
An Introduction to Statistical Learning

James, Witten, Hastie & Tibshirani

August 13, 2026

An Introduction to Statistical Learning with Applications in R

a.k.a. ISLR



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Visit <https://www.statlearning.com/>.

Plan

Simple Linear Regression

- Estimating the Coefficients

- Assessing the Accuracy of the Coefficient Estimates

- Assessing the Accuracy of the Model

Multiple Linear Regression

- Estimating the Regression Coefficients

- Some Important Questions

Other Considerations in the Regression Model

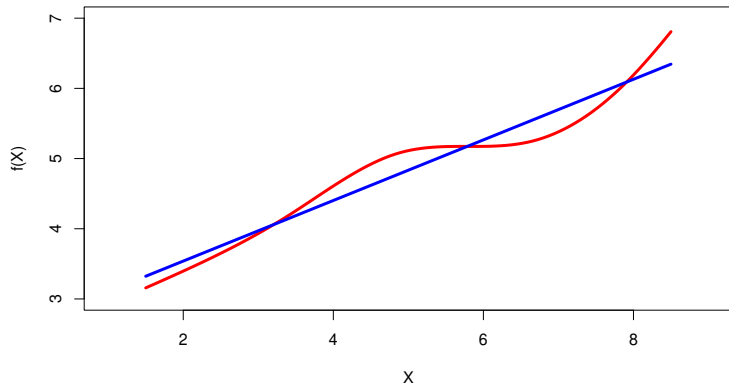
- Qualitative Predictors

- Extensions of the Linear Model

- Potential Problems

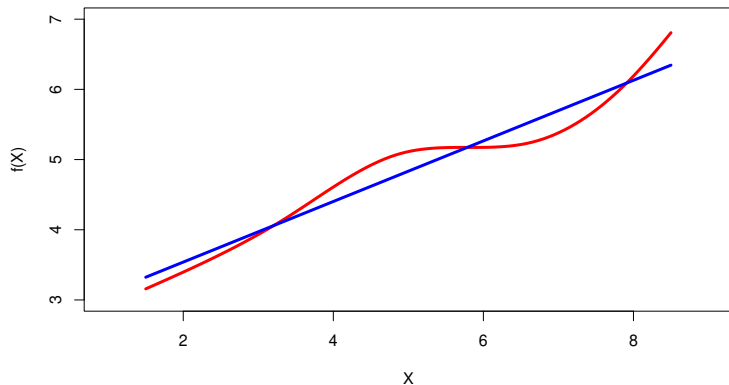
Linear Regression

- ▶ Linear regression is a simple approach to supervised learning. It assumes that the dependence of Y on $X_1, X_2, \dots, X_p$ is linear.
- ▶ True regression functions are never linear!



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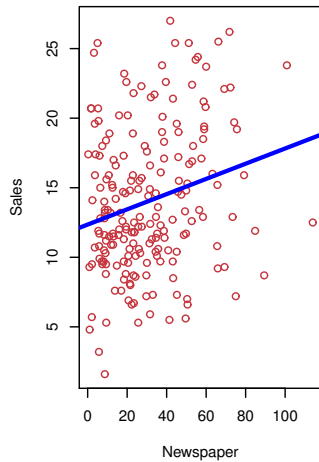
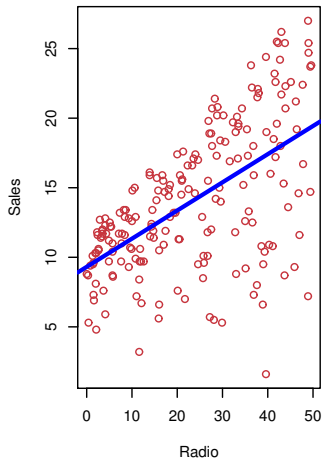
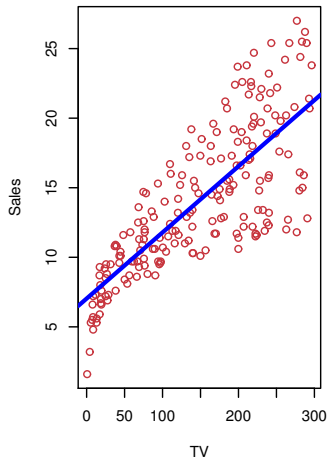
- ▶ Although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

Linear Regression for the Advertising Data

Consider the advertising data. Questions we might ask:

- ▶ Is there a relationship between advertising budget and sales?
- ▶ How strong is the relationship between advertising budget and sales?
- ▶ Which media contribute to sales?
- ▶ How accurately can we predict future sales?
- ▶ Is the relationship linear?
- ▶ Is there synergy among the advertising media?

Advertising data



Simple Linear Regression

Simple linear regression using a single predictor X :

- ▶ We assume a model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where β_0 and β_1 are two unknown constants that represent the **intercept** and **slope**, also known as **coefficients** or **parameters**, and ϵ is the error term.

- ▶ Given some estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model coefficients, we predict future sales using:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where $\hat{y}$ indicates a prediction of Y on the basis of $X = x$. The **hat** symbol denotes an estimated value.

Estimation of the Parameters by Least Squares

- ▶ Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction for Y based on the i -th value of X . Then $e_i = y_i - \hat{y}_i$ represents the i -th **residual**.
- ▶ We define the **residual sum of squares** (RSS) as:

$$RSS = e_1^2 + e_2^2 + \cdots + e_n^2$$

or equivalently as:

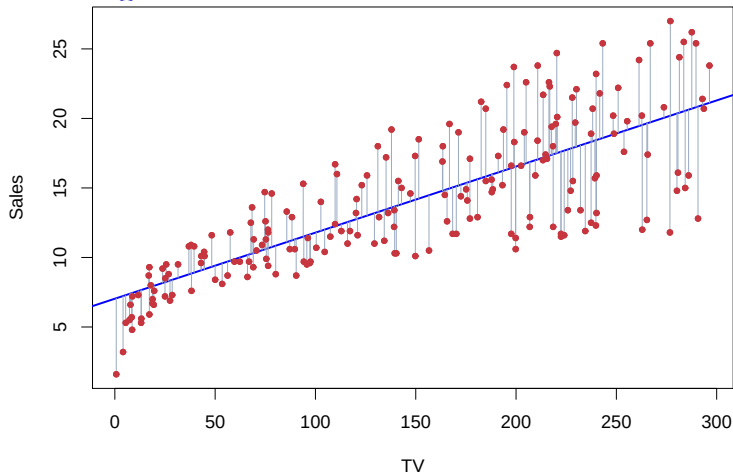
$$RSS = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \cdots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

- ▶ The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS. The minimizing values are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{y} \equiv \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$ are the sample means.

Example: advertising data



The least squares fit for the regression of **sales** and **TV**. In this case a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.

Assessing the Accuracy of the Coefficient Estimates

- The standard error of an estimator reflects how it varies under repeated sampling:

$$SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

where $\sigma^2 = \text{Var}(\epsilon)$.

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where $\sigma^2 = \text{Var}(\epsilon)$.

- ▶ These standard errors can be used to compute **confidence intervals**. A 95% confidence interval is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter. It has the form:

$$\hat{\beta}_1 \pm 2 \cdot SE(\hat{\beta}_1)$$

Confidence Intervals Continued

- That is, there is approximately a 95% chance that the interval:

$$[\hat{\beta}_1 - 2 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot SE(\hat{\beta}_1)]$$

will contain the true value of β_1 (under a scenario where we got repeated samples like the present sample).

Confidence Intervals Continued

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- ▶ For the advertising data, the 95% confidence interval for β_1 is $[0.042, 0.053]$.

Hypothesis Testing

- Standard errors can also be used to perform **hypothesis tests** on the coefficients. The most common hypothesis test involves testing the **null hypothesis** of:

H_0 : There is no relationship between X and Y
versus the **alternative hypothesis**

H_A : There is some relationship between X and Y .

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versus the **alternative hypothesis**

H_A : There is some relationship between X and Y .

- Mathematically, this corresponds to testing:

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0$$

since if $\beta_1 = 0$ then the model reduces to $Y = \beta_0 + \epsilon$, and X is not associated with Y .

Hypothesis Testing – continued

- ▶ To test the null hypothesis, we compute a *t-statistic*, given by:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

- ▶ This will have a *t*-distribution with $n - 2$ degrees of freedom, assuming $\beta_1 = 0$.
- ▶ Using statistical software, it is easy to compute the probability of observing any value equal to $|t|$ or larger. We call this probability the *p-value*.

Results for the Advertising Data

	Coefficient	Std. Error	<i>t</i>-statistic	<i>p</i>-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Assessing the Overall Accuracy of the Model

- We compute the **Residual Standard Error**:

$$RSE = \sqrt{\frac{1}{n-2}RSS} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

where the **residual sum-of-squares** is $RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.

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- **R-squared** or fraction of variance explained is:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

where $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ is the **total sum of squares**.

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- In simple linear regression, $R^2 = r^2$, where r is the sample correlation between X and Y :

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Advertising Data Results

Quantity	Value
Residual Standard Error	3.26
R^2	0.612
F -statistic	312.1

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Multiple Linear Regression

- Estimating the Regression Coefficients

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Other Considerations in the Regression Model

- Qualitative Predictors

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Multiple Linear Regression

- ▶ Here our model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

- ▶ We interpret β_j as the **average** effect on Y of a one unit increase in X_j , **holding all other predictors fixed**. In the advertising example, the model becomes:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon$$

Interpreting Regression Coefficients

- ▶ The ideal scenario is when the predictors are uncorrelated – a **balanced design**:
 - ▶ Each coefficient can be estimated and tested separately.
 - ▶ Interpretations such as “*a unit change in X_j is associated with a β_j change in Y , while all other variables stay fixed*”, are possible.
- ▶ Correlations amongst predictors cause problems:
 - ▶ The variance of all coefficients tends to increase, sometimes dramatically.
 - ▶ Interpretations become hazardous: when X_j changes, everything else changes.
- ▶ **Claims of causality** should be avoided for observational data.

The Woes of (Interpreting) Regression Coefficients

“Data Analysis and Regression” – Mosteller and Tukey (1977)

A regression coefficient β_j estimates the expected change in Y per unit change in X_j , **with all other predictors held fixed**. But predictors usually change together!

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- ▶ **Example 2:** Y = number of tackles by a football player in a season; W and H are his weight and height. Fitted regression model is $\hat{Y} = b_0 + 0.50W - 0.10H$. How do we interpret $\hat{\beta}_2 < 0$?

Two Quotes by Famous Statisticians

George Box

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Two Quotes by Famous Statisticians

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“Essentially, all models are wrong, but some are useful.”

Fred Mosteller and John Tukey (paraphrasing George Box)

“The only way to find out what will happen when a complex system is disturbed is to disturb the system, not merely to observe it passively.”

Estimation and Prediction for Multiple Regression

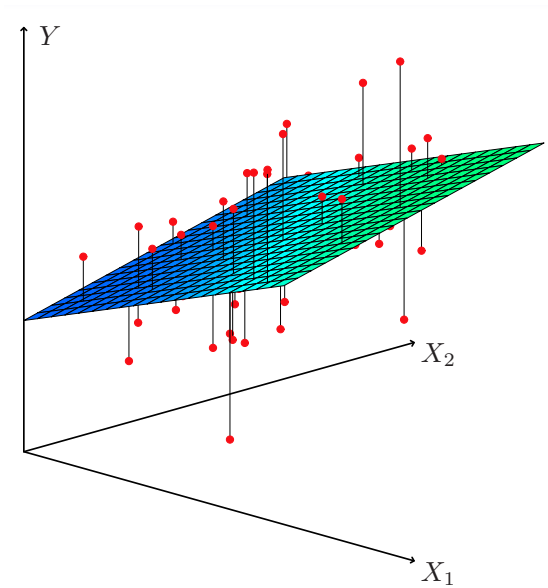
- ▶ Given estimates $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$, we can make prediction using the formula:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p$$

- ▶ We estimate $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ as the values that minimize the sum of squared residuals:

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2$$

This is done using standard statistical software. The values $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ that minimize RSS are the multiple least squares regression estimates.



Results for Advertising Data

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	2.939	0.3119	9.42	<0.0001
TV	0.046	0.0014	32.81	<0.0001
radio	0.189	0.0086	21.89	<0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Correlations:

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Some Important Questions

1. Is at least one of the predictors $X_1, X_2, \dots, X_p$ useful in predicting the response?

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Is at Least One Predictor Useful?

For the first question, we can use the F -statistic:

$$F = \frac{\frac{(TSS-RSS)}{p}}{\frac{RSS}{n-p-1}} \sim F_{p, n-p-1}$$

Quantity	Value
Residual Standard Error	1.69
R^2	0.897
F -statistic	570

Deciding on the Important Variables

- ▶ **All Subsets / Best Subsets Regression:** Compute least squares fit for all possible subsets and choose based on a criterion balancing training error and model size.

Deciding on the Important Variables

- ▶ **All Subsets / Best Subsets Regression:** Compute least squares fit for all possible subsets and choose based on a criterion balancing training error and model size.
- ▶ However, there are 2^p possible models (e.g., $p = 40 \implies > 1$ billion models!).
- ▶ Instead we need an automated approach that searches through a subset of them. The book discusses two commonly use approaches next.

Variable Selection Methods

Forward Selection

- ▶ Begin with the **null model** (a model that contains an intercept but no predictors).
- ▶ Fit p simple linear regression and add to the null model the variable that results in the lowest RSS.
- ▶ Add to that model the variable that results in the lowest RSS amongst all two-variable models.
- ▶ Continue adding variables one by one until a stopping rule is satisfied. i.e. when all remaining variables have a *p-value* above some threshold.

Variable Selection Methods

Backward Selection

- ▶ Start with all variables in the model.
- ▶ Remove the variable with the largest *p-value* (that is, the variable that is the least statistically significant).
- ▶ The new $(p - 1)$ -variable model is fit, and the variable with the largest *p-value* is removed.
- ▶ Continue until a stopping rule is reached. i.e. we may stop when all remaining variable have a significant *p-value* defined by some significance threshold.

Model Selection Criteria

Other systematic criteria for choosing an optimal model path include:

- ▶ Mallow's C_p
- ▶ Akaike Information Criterion (AIC)
- ▶ Bayesian Information Criterion (BIC)
- ▶ Adjusted R^2
- ▶ Cross-Validation (CV)

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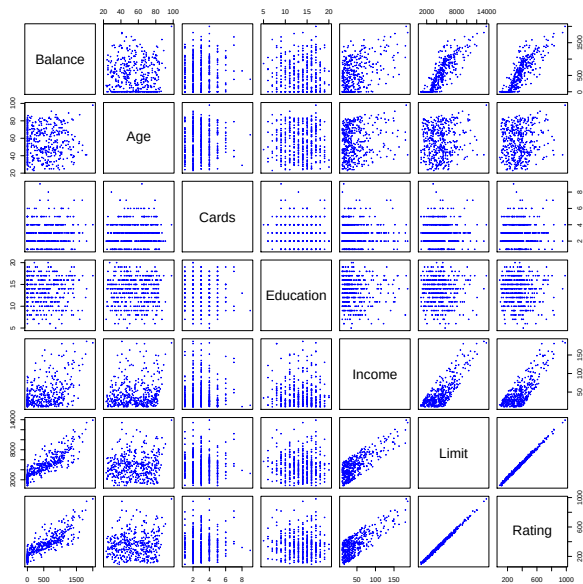
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Qualitative Predictors

- ▶ Qualitative predictors take a discrete set of values (also known as **categorical** predictors or **factor variables**).
- ▶ i.e. In addition to the 7 quantitative variables shown, there are four qualitative variables: **gender**, **student** (student status), **status** (marital status), and **ethnicity** (Caucasian, African American or Asian).

Credit card data



Regression

Qualitative Predictors –continued

- ▶ Example: investigate differences in credit card balance between males and females, ignoring the other variables.
- ▶ We create a new variable (a binary predictor, e.g. gender: Female/Male):

$$x_i = \begin{cases} 1 & \text{if } i\text{-th person is female} \\ 0 & \text{if } i\text{-th person is male} \end{cases}$$

- ▶ Resulting model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if female} \\ \beta_0 + \epsilon_i & \text{if male} \end{cases}$$

Qualitative Predictors –continued

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- ▶ Interpretation?

Credit card data –continued

Results for gender model:

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	509.80	33.13	15.389	<0.0001
gender[Female]	19.73	46.05	0.429	0.6690

Qualitative Predictors with More Than Two Levels

- ▶ With $k > 2$ levels, create $k - 1$ dummy variables.
- ▶ i.e. for **ethnicity** (Asian, Caucasian, African American) create two dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if Asian} \\ 0 & \text{otherwise} \end{cases}, \quad x_{i2} = \begin{cases} 1 & \text{if Caucasian} \\ 0 & \text{otherwise} \end{cases}$$

- ▶ Resulting model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if Asian} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if Caucasian} \\ \beta_0 + \epsilon_i & \text{if African American (baseline)} \end{cases}$$

Qualitative Predictors with More Than Two Levels –continued

- Then both of these variables can be used in the regression equation, in order to obtain the model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if Asian} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if Caucasian} \\ \beta_0 + \epsilon_i & \text{if African American (baseline)} \end{cases}$$

Note: There will always be one fewer dummy variable than the number of levels. The level with no dummy variable –AA in this example– is known as the **baseline**

Results for ethnicity

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	531.00	46.32	11.464	<0.0001
ethnicity[Asian]	-18.69	65.02	-0.287	0.7740
ethnicity[Caucasian]	-12.50	56.68	-0.221	0.8260

Extensions of the Linear Model

Removing the additive assumption: *interactions* and *nonlinearity*...

► Interactions:

- In our previous analysis of the **Advertising** data, we assumed that the effect on **sales** of increasing one advertising medium is independent of the amount spent on the other media.
- i.e. the linear model:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper}$$

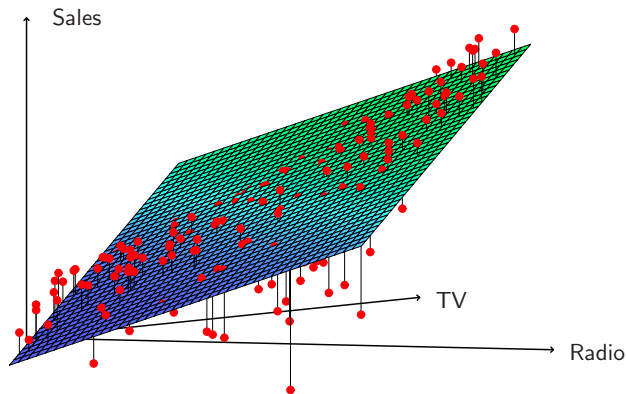
states that the average effect on **sales** of a one-unit increase in TV is always β_1 , regardless of the amount spent on **radio**.

Extensions of the Linear Model –continued

► Interactions:

- But suppose that spending money on radio advertising actually increases the effectiveness of TV advertising, so that the slope term for TV should increase as **radio** increases.
- In this situation, given a fixed budget of \$100,000, spending half on **radio** and half on TV may increase **sales** more than allocating the entire amount to either TV or to **radio**.
- In marketing, this is known as a **synergy** effect, and in statistics it is referred to as an **interaction** effect.

Interaction in the Advertising data?



- ▶ When levels of either TV or radio are low, then the true **sales** are lower than predicted by the linear model.
- ▶ But when advertising is split between the two media, then the model tends to underestimate **sales**.

Modelling Interactions – Advertising Data Results

Model takes the form:

$$\begin{aligned}\text{sales} &= \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times (\text{TV} \times \text{radio}) + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \text{radio}) \times \text{TV} + \beta_2 \times \text{radio} + \epsilon\end{aligned}$$

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	6.7502	0.2480	27.23	< 0.0001
TV	0.0191	0.0015	12.70	< 0.0001
radio	0.0289	0.0089	3.24	0.0014
TV×radio	0.0011	0.0001	20.73	< 0.0001

Interpretation

- ▶ The results in this table suggests that interactions are important.
- ▶ The *p-value* for the interaction term $\text{TV} \times \text{radio}$ is extremely low, indicating that there is strong evidence for $H_A : \beta_3 \neq 0$.
- ▶ R^2 increases from 89.7% (additive) to 96.8% (with interaction).

Interpretation –continued

- ▶ Explains variability remaining after fitting additive model!
- ▶ This means that

$$\frac{96.8 - 89.7}{100 - 89.7} = 69\%$$

of the variability in **sales** that remains after fitting the additive model has been explained by the interaction term.

- ▶ The coefficient estimates in the table suggest that an increase in TV advertising of \$1,000 is associated with increased sales of

$$(\hat{\beta}_1 + \hat{\beta}_3 \times \mathbf{radio}) \times 1000 = 19 + 1.1 \times \mathbf{radio} \text{ units.}$$

- ▶ An increase in radio advertising of \$1,000 will be associated with an increase in sales of

$$(\hat{\beta}_1 + \hat{\beta}_3 \times \mathbf{TV}) \times 1000 = 29 + 1.1 \times \mathbf{TV} \text{ units.}$$

The Hierarchy Principle

- Sometimes it is the case that an interaction term has a very small p -value, but the associated main effects (in this case, `TV` and `radio`) do not.

Hierarchy Principle

If we include an interaction in a model, we should also include the main effects, even if the p -values associated with their coefficients are not significant.

The Hierarchy Principle –continued

- ▶ The rationale for this principle is that interactions are hard to interpret in a model without main effects –their meaning is changed.
- ▶ Specifically, the interaction terms also contain main effects, if the model has no main effect terms.

Interactions between qualitative and quantitative variables

- ▶ Consider the **Credit** data set, and suppose that we wish to predict **balance** using **income** (quantitative) and **student** (qualitative).
- ▶ Without an interaction term, the model takes the form:

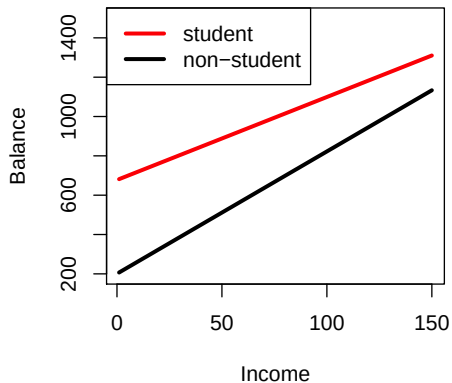
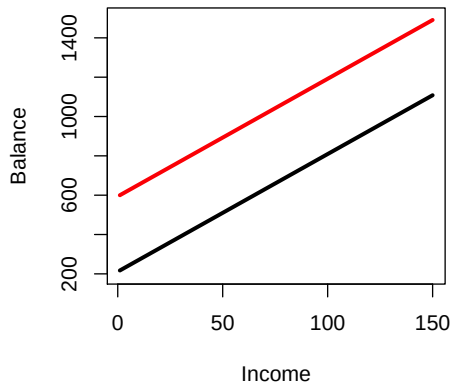
$$\begin{aligned}\text{balance}_i &\approx \beta_0 + \beta_1 \times \text{income}_i + \begin{cases} \beta_2 & \text{if } i\text{-th person is a student} \\ 0 & \text{if } i\text{-th person is not a student} \end{cases} \\ &= \beta_{\times} \text{income}_i + \begin{cases} \beta_0 + \beta_2 & \text{if } i\text{-th person is a student} \\ \beta_0 & \text{if } i\text{-th person is not a student} \end{cases}\end{aligned}$$

Interactions between qualitative and quantitative variables –continued

- With interactions, it takes the form:

$$\begin{aligned}\text{balance}_i &\approx \beta_0 + \beta_1 \times \text{income}_i + \begin{cases} \beta_2 + \beta_3 \times \text{income}_i & \text{if student} \\ 0 & \text{if not student} \end{cases} \\ &= \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \times \text{income}_i & \text{if student} \\ \beta_0 + \beta_1 \times \text{income}_i & \text{if not student} \end{cases}\end{aligned}$$

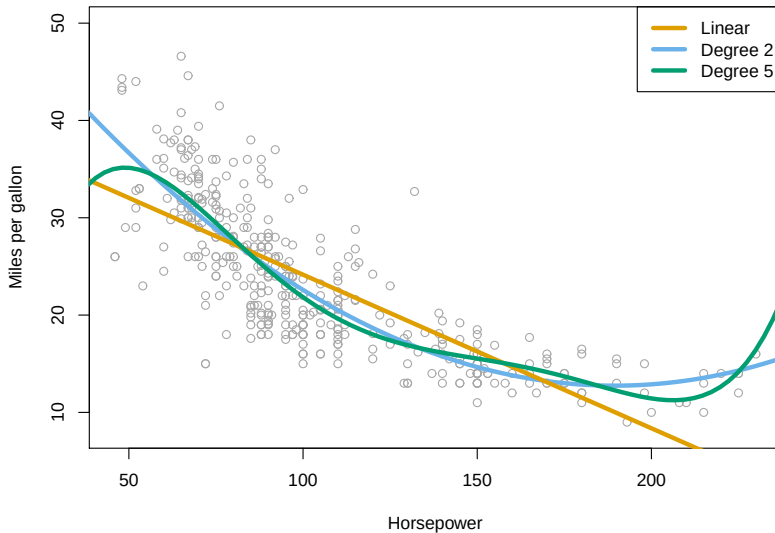
Interactions between qualitative and quantitative variables –continued



- Credit data; Left: no interaction between `income` and `student`. Right: with an interaction term between `income` and `student`.

Non-linear Effects of Predictors

polynomial regression on Auto data



Non-linear Effects of Predictors –continued

- ▶ Polynomial Regression extends linear models to non-linear relationships.
- ▶ The figure suggests that:

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 + \epsilon$$

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	56.9001	1.8004	31.6	<0.0001
horsepower	-0.4662	0.0311	-15.0	<0.0001
horsepower ²	0.0012	0.0001	10.1	<0.0001

What We Did Not Cover & Generalizations

Potential Issues in Regression (Section 3.3.3)

Outliers, Non-constant variance of error terms, High leverage points, Collinearity.

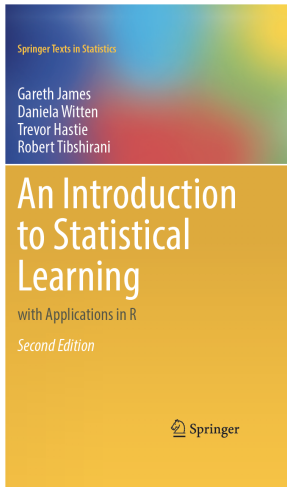
Generalizations of the Linear Model

- ▶ **Classification:** Logistic regression, Support Vector Machines.
- ▶ **Non-linearity:** Kernel smoothing, Splines, GAMs, Nearest Neighbors.
- ▶ **Interactions:** Tree-based methods, Bagging, Random Forests, Boosting (these also capture non-linearities).
- ▶ **Regularized fitting:** Ridge regression, Lasso

End of Chapter 3.

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a.k.a. ISLR



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Regression